A Support–Faithfulness Contrast Aligns Across Model Families up to a Linear Map but Does Not Transfer to Real Hallucinations: A Matched-Pair Study

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Abstract

Internals-based, reference-free hallucination detection treats *context-faithfulness*—is an answer entailed by the provided context?—as a linear direction in the residual stream. But in the standard extractive setup, faithfulness is almost perfectly correlated with *lexical overlap* ($r{=}0.99$ in our data), so an apparent "grounding probe" may be a string-matching detector. We introduce a matched SUPPORT×OVERLAP construction that *partially* decouples the two (halving the confound to $r{=}0.50$), with a wrong-question trap that fools an off-the-shelf NLI model. Across four instruction-tuned 7–9B families we find, at each step, that an apparent grounding signal is something else once a confound is removed. (i) Within the overlap-controlled stratum a support contrast is decodable, but a *no-context ablation* shows it is mostly question–answer compatibility—not use of the context—and is inseparable from factoid-correctness by construction, so we claim no context-grounding signal. (ii) The contrast aligns across families up to a linear map, but answer-length, overlap and factuality directions align just as well, so the alignment is a generic matched-stimulus property, not grounding-specific. (iii) The synthetic direction does not transfer to real hallucinations: on RAGTruth it scores below NLI, confidence, and lexical overlap, and only an in-domain probe trained on human labels recovers a usable signal. Our contribution is a confound-controlled methodology, a released wrong-question diagnostic, and a cautionary result: internals-based grounding claims survive one confound only to fall to the next. All experiments are API-free and run on a single A100-40G.

1 Introduction

A grounded language model should answer from the *provided context*, not from its parametric memory or from surface cues. Reference-free, internals-based methods operationalize this by reading a linear direction from the residual stream: much prior work decodes a "truth" or "lying" direction (Azaria and Mitchell, 2023; Marks and Tegmark, 2024; Bürger et al., 2024), and more recent work extends this style of analysis to a *contextual* faithfulness direction (O'Neill et al., 2025)—an attractive recipe because it needs no external judge and costs roughly one forward pass. Two distinct notions are routinely conflated under this banner: *parametric-factuality*—is a claim true in the model's memory?—and *context-faithfulness*—is a claim entailed by the given context? Recent work treats both as activation objects and asks how they relate (Wang et al., 2025), but samples the two notions from *different* datasets, so any measured geometry is confounded with topic and lexicon.

We identify a sharper confound that undermines the faithfulness side specifically. In the standard extractive QA setup used to build such probes (e.g. entity-swap contexts (Longpre et al., 2021)), an answer is context-faithful *iff* the answer string occurs in the context. Faithfulness is therefore almost perfectly correlated with *lexical overlap* ($r{=}0.99$ in our data), and a probe that appears to read "grounding" may simply detect string copying. Indeed, after regressing out lexical overlap, an apparently excellent grounding probe collapses to chance.

A matched control. We introduce a SUPPORT×OVERLAP construction (§3) that crosses semantic support with lexical presence in a balanced 2×2. Two cells are model-free: a *wrong-question trap*, where the answer is present but answers a different question (supported=no, overlap=yes), and the usual unsupported-absent cell; a paraphrase cell closes the design. This drops $\mathrm{corr}(\text{SUPPORT}, \text{OVERLAP})$ from 0.99 to 0.50, and—tellingly—an off-the-shelf NLI model is fooled by the clean wrong-question trap $\sim$25% of the time, confirming that lexical overlap misleads even "gold" entailment checks.

Contributions. With this control we run a careful study across four model families (Llama-3.1-8B, Qwen2.5-7B-Instruct, Mistral-7B-v0.3, Gemma-2-9B). The result is a methodology and a chain of confound-controlled negatives: each apparent "grounding" signal, once a confound is removed, turns out to be something else.

1. **An overlap-controlled internal signal that is *not* context-grounding (§5.1).** Within the answer-present stratum (lexical overlap held constant), a support contrast is decodable on all four families (collision-free raw AUROC near-ceiling, $\approx$0.99; 0.76–0.88 after purging model confidence). But a *no-context ablation* shows it is mostly question–answer compatibility: removing the context barely changes it (raw 0.95–0.98, confidence-purged unchanged), so the probe separates the matched answer from the wrong-question answer *without reading the context*. Together with the fact that support coincides with our *synthetic* factuality label here by construction, this shows an overlap-controlled internal contrast can still be compatibility and factoid-correctness—we therefore do *not* claim a context-grounding signal, even an overlap-controlled one.

2. **Cross-family alignment up to a linear map (§5.3).** An orthogonal-Procrustes map carries the support direction between families so the transported direction is nearly collinear with the target family's *native* direction (cosine $\approx$0.98, vs. $\approx$0.01 for a random direction through the same map), whereas an anchor-projection transfer reaches only 0.67 AUROC (Kim and Han, 2026). The transferred-probe AUROC (0.91) equals the within-family bound but is ceiling-saturated, and signed map-aware nulls sit at chance (0.51); we therefore rest the claim on the cosine. Factuality, answer-length and overlap directions transport identically while a confidence direction does not, indicating a shared stimulus-derived contrast rather than a grounding-specific axis; we claim no universality beyond these four 7–9B models (Puri et al., 2025).

3. **No confidence asymmetry (§5.2).** Contrary to a natural hypothesis, softmax confidence and activation norm do not absorb the factuality axis more than the faithfulness axis; the effect is null.

4. **Synthetic axes do not transfer to real hallucinations (§5.4).** Applied to RAGTruth (Niu et al., 2024), the controlled-stimulus support direction scores *below* NLI, confidence, and lexical overlap among single-pass API-free baselines; within a task no internals- or NLI-based signal—including a sentence-level SummaC-style NLI baseline—exceeds $\approx$0.6 AUROC, and even lexical overlap, the strongest baseline, reaches only macro 0.647. An in-domain probe trained on RAGTruth human labels recovers a usable signal (0.77–0.82 on the three RAGTruth families; exceeds a conservative single-hypothesis NLI baseline but ties lexical overlap on Qwen), so a real support signal exists in the internals but is a *different direction* from the synthetic axis and needs in-domain annotation.

The takeaway for the "faithful and efficient" agenda is cautionary but useful: internals-based support contrasts identified on controlled data are real and remarkably portable across model families (though not grounding-specific), yet are *not*, by themselves, deployable hallucination detectors—on real data, lexical overlap still dominates.

2 Related Work

Geometry of truth and faithfulness. Linear "truth" directions are decodable from activations (Azaria and Mitchell, 2023; Marks and Tegmark, 2024) and partly shared across models (Bürger et al., 2024), but are task-dependent and often fail to transfer across tasks (Azizian et al., 2025; Bao et al., 2025). Wang et al. (2025) treat factuality and faithfulness hallucinations as activation subspaces and argue they overlap; crucially they draw the two notions from different corpora, leaving the geometry confounded with surface form. Adarsh et al. (2026) study how a single truth vector rotates when context is added. We instead hold the two notions on *matched* stimuli and control lexical overlap explicitly; we note, however, that our factuality label is a synthetic gold-answer relabeling rather than a parametric memory-truth probe, so we engage this line methodologically without claiming to adjudicate whether factuality and faithfulness share a direction.

Knowledge conflict and source attribution. Probes can detect knowledge conflict and predict which source a model follows (Zhao et al., 2024; Sun et al., 2025; Brink et al., 2026). These are behavioral/source signals; we ask whether a *content* grounding axis exists that is not reducible to lexical overlap.

Cross-model transfer. Linear concept directions can be aligned across models via orthogonal Procrustes (Schönemann, 1966; Puri et al., 2025) or anchor projection (Kim and Han, 2026). We apply both to a support direction; Procrustes renders it nearly collinear with the target's native direction (cosine ≈0.98) whereas anchor projection does not.

Reference-free hallucination detection. Reference-free detectors fall into three families: *external* entailment/alignment scorers (Zha et al., 2023); *internals*-based probes—single residual directions (O'Neill et al., 2025) and the truth/faithfulness directions above; and *confidence*/correctness signals (Cho et al., 2026; Moreno Cencerrado et al., 2025), with faithfulness-aware UQ (Fadeeva et al., 2025) combining several. We use one baseline from each family on RAGTruth—NLI (external), model confidence, and lexical overlap—and find our controlled-stimulus axis does not beat them.

3 The SUPPORT×OVERLAP Construction

We build matched statements from NQ-Swap (Longpre et al., 2021), which provides, per question q, a gold answer a_{true} supported by an original context c_{org} and a substituted entity a_{swap} supported by an entity-swapped context c_{sub}. We define two bits per (context, answer): SUPPORT S (does the given context entail the asserted answer for the asked question?) and OVERLAP O (does the answer string occur in the context?). The 2×2 design has four cells, of which we realize three (cell B is unrealized in this release; see below):

cell	(question, context, answer)	S	O
A	$(q,\ c_{\text{org}},\ a_{\text{true}})$	1	1
B	$(q,\ \text{paraphrase}(c_{\text{org}}),\ a_{\text{true}})$	1	0
C	$(q',\ c_{\text{org}},\ a_{\text{true}})$	0	1
D	$(q,\ c_{\text{sub}},\ a_{\text{true}})$	0	0

Cell C is a *wrong-question trap*: a_{true} is present in c_{org} but is offered as the answer to a different question q' (the next item's question). Cell B paraphrases c_{org} so the answer is entailed but not copied (open instruct model, kept only if token-overlap≤0.34 and an NLI model still entails it); in this release cell B has *zero* yield, so all reported analyses use the model-free cells A/C/D with $\text{corr}(S, O)=0.50$ (0.57 after removing the donor-collision items below; an orthogonalized corr≈0 design would require cell B). One caveat in cell C: NQ-Swap dev has only 418 distinct questions over 1,200 items, so for 278 items (23.2%) the donor q' is string-identical to q and a_{true} *does* answer it; we flag these *donor-collision* items and report cell-C results with them removed (the full construction will dedup $q' \neq q$). On *clean* cell C an open NLI model (He et al., 2023) still rates 25.5% of unsupported-but-present items as "supported" (mean entailment 0.31, vs. 0.74 for cell A and 0.18 for cell D); the raw 37.6% rate is inflated by the collisions, which NLI entails 77.7% of the time (correctly). Lexical overlap thus misleads "gold" entailment even after the label error is purged (Fig. 1).

4 Probing, Separability, and Transfer

Directions. For a (model, layer) we extract last-token residual states X over the asserted answer span (one forward pass, bf16, left-padded). A SUPPORT direction d_S and FACTUALITY direction d_F are mass-mean (difference-of-means) estimates (Marks and Tegmark, 2024); signed projections Xd are the probe scores. We report layer-wise cross-validated AUROC under *GroupKFold on* ITEM_ID: because cells A/C/D of an item reuse the same context, an ungrouped split would leak near-duplicate rows across folds, so all statements of one item are kept in a single fold. Axis polarity is fixed on the *train* fold; the test projection is scored as-is (no $\max(\text{AUC}, 1-\text{AUC})$ flip), so a non-predictive direction correctly averages to 0.5 rather than being floored above it. All point estimates carry item cluster-bootstrap 95% CIs.

Overlap control. To test grounding *beyond* overlap we (i) restrict to the answer-present stratum $O=1$ (cells A vs. C), where lexical presence is constant, and (ii) residualize X on surface covariates (overlap, length) before re-probing.

Null-calibrated separability. We treat each notion as an undirected axis and measure the acute angle $\angle(d_S, d_F)$. Significance is assessed against a within-notion null built by splitting one notion's data in half and measuring the half-vs-half angle; a permutation test with n_{null}=10,000 draws gives p. We report the resolution bound $p<1/n_{\text{null}}=10^{-4}$ rather than $p=0$ when no null draw exceeds the observed angle, and state that the test is run per family at the a-priori mid layer (four families ×

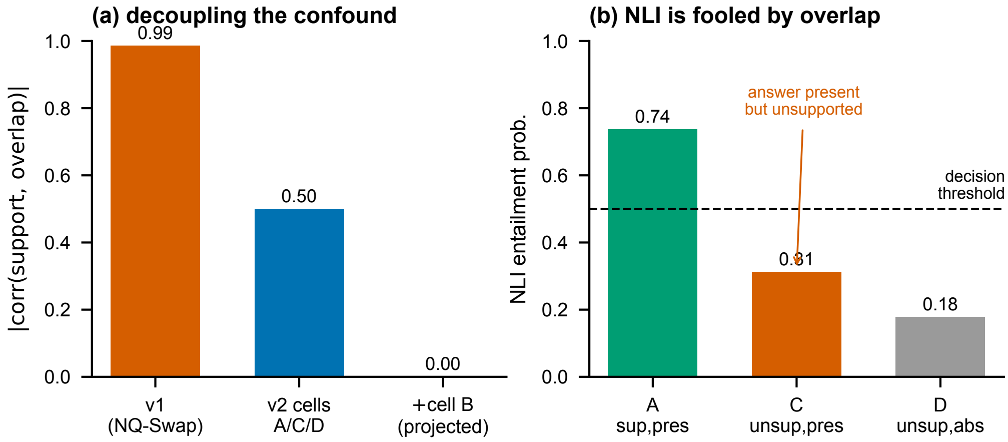

Figure 1: **Partially decoupling support from lexical overlap.** (a) The matched construction drops $|\mathrm{corr}(\textsc{support}, \textsc{overlap})|$ from 0.99 (extractive NQ-Swap) to 0.50 (model-free cells A/C/D), and would reach ≈ 0 with the paraphrase cell B (projected; cell B has zero yield in this release). (b) Mean NLI entailment by *clean* cell: on the wrong-question trap (C), where the answer is present but unsupported, NLI is still pulled toward "supported."

one layer).

Confidence purge. We regress each hidden dimension on confidence covariates (max-softmax, answer log-prob, activation norm) and re-probe the residual, reporting the fraction of probe signal absorbed per axis.

Cross-family transfer. Families differ in hidden size, so we fit a semi-orthogonal Procrustes map between two models' residual spaces on a shared calibration split, transport a source-trained probe, and measure target AUROC. The calibration/evaluation split is a *GroupKFold on* ITEM_ID (items in the calibration set are disjoint from the evaluation set), so the map is never fit and tested on near-duplicate rows of the same item; transported-probe polarity is fixed on the target calibration fold. We compare to an unconstrained ridge map, the anchor-projection ACS baseline (Kim and Han, 2026), a within-family upper bound, and a random-direction floor (also scored without the polarity flip, so it sits at chance rather than being inflated above 0.5).

5 Experiments

Setup. Four instruction-tuned families: Llama-3.1-8B (Grattafiori et al., 2024), Qwen2.5-7B (Qwen Team et al., 2024), Mistral-7B-v0.3 (Jiang et al., 2023), Gemma-2-9B (Gemma Team et al., 2024). CtrlPairs: 1,200 NQ-Swap items → 3,593 matched statements (cells A/C/D). RAGTruth: 2,000 test responses with response-level hallucination labels (Niu et al., 2024). Baselines: NLI (the `DeBERTa-v3-base-mnli-fever-anli` checkpoint (Laurer et al., 2024)—DeBERTaV3 (He et al., 2023) fine-tuned on MNLI, FEVER and ANLI (Williams et al., 2018; Thorne et al., 2018; Nie et al., 2020)) in two variants—(i) *whole-response*: a single (context, declarativized-answer) entailment query (deliberately simple and conservative for multi-sentence responses); and (ii) *sentence-level*, a SummaC-style decomposition (Laban et al., 2022) that entailment-checks each response sentence against every context sentence, takes the max over context, and aggregates over response sentences by mean (SummaC-ZS) or min (for tractability we cap context at 60 sentences and pairs at 256 tokens, which can only lower SummaC and is thus conservative)—plus max-softmax confidence and lexical overlap. Everything is API-free; the NLI baselines use the small DeBERTa model and all activation extraction runs on one A100-40G.

5.1 The within-stratum signal is question–answer compatibility, not grounding

Leakage-controlled protocol. Cells A/C/D of an item share the original context c_{org} verbatim (A and C are the same context with a different question), so a plain row-level split would place near-duplicate rows of one item on both sides of the train/test boundary. We therefore use *GroupKFold on* ITEM_ID throughout: all statements derived from one NQ-Swap item fall in the same fold, for both the C1 cross-validation and the Procrustes calibration/evaluation split (§5.3). We resolve each estimated axis's sign on the *train* fold only (never on the test fold), so test AUROC can fall below 0.5 and is not floored by a $\max(\mathrm{AUC}, 1{-}\mathrm{AUC})$ flip. To avoid selecting the layer on the very number we report, we fix the analysis layer *a priori* at relative depth 0.5 (the nearest cached layer to $0.5\,L_{\mathrm{tot}}$: L=14/16/16/21 for Qwen/Mistral/Llama/Gemma), matching the relative-depth convention already used for C3/C4 and for Fig. 2; we additionally report the mean±sd across the mid-layer band (0.4–0.6 depth) to show the result is not a single-layer artifact. Confidence intervals are item cluster bootstraps (resampling ITEM_ID with replacement).

Table 1 reports support decodability within the answer-present stratum O=1 (cells A vs. C), where lexical overlap is held constant, on the *collision-free* cells: we drop the 23.2% of cell-C items whose donor question is string-identical to the original (§3), which would otherwise mislabel genuinely-supported items as unsupported. Three observations bound the interpretation. First, the raw within-stratum AUROC is *near-ceiling* (0.99; 0.988–0.999 across families): once mislabels are removed, A and C are almost perfectly separable. We read this less as strong grounding than as evidence that the contrast is *easy*—within O=1 it varies *only the question* (context and answer are fixed), so a probe can in principle separate A from C by question–answer appropriateness without using the context. Second, the operative confound is *model confidence* (overlap and answer length are uninformative here, AUROC 0.500): answer log-prob and NLI separate A from C on their own, so we residualize the hidden states on log-prob and max-softmax; the support contrast then falls to 0.76–0.88, which we take as the primary confound-controlled estimate. Third—and decisively—a *no-context ablation* settles what the signal is: re-extracting the A/C activations with the context *removed* (question+answer only) leaves $\mathrm{supp}_{O=1}$ almost unchanged (raw 0.95–0.98 vs. 0.99 with context; the confidence-purged estimate does not collapse either, 0.80–0.83, rising for two families and falling only ∼0.08 for Mistral), and replacing the context with an unrelated passage gives the same (0.94–0.98). The within-O=1 contrast is therefore mostly *question–answer compatibility*: the probe separates A (whose question the answer fits) from C (whose donor question it does not) *without reading the context at all*. A cross-strata control (train A vs. D, which fixes the question but also varies overlap since D is answer-absent; evaluate A vs. C) does transfer at 0.66–0.88, so *some* context-sensitive component exists, but it is an upper bound and not separable from the dominant compatibility signal here. Finally, support and our *synthetic* factuality label are *identical* within O=1 by construction (they diverge only in the answer-absent cell D). We therefore do *not* claim a context-grounding signal: within the overlap-controlled stratum the decodable contrast is question–answer compatibility—not isolated from factoid-correctness, and not shown to use the context at all.

model	L	$\mathrm{supp}_{O=1}$	+conf	cross
Qwen2.5	14	0.988	0.757	0.660
Mistral	16	0.996	0.878	0.870
Llama	16	0.997	0.784	0.876
Gemma	21	0.999	0.806	0.827

Table 1: **C1 (confound-controlled, collision-free).** Layer fixed a priori at relative depth 0.5; GroupKFold on ITEM_ID; polarity resolved on the train fold; donor-collision cell-C mislabels removed (§3; n_A=1200, n_C=922). $\mathrm{supp}_{O=1}$ is grouped-CV support AUROC within the answer-present stratum—near-ceiling once mislabels are removed, because the contrast varies only the question and is easily separable. *+conf*: after residualizing the hidden states on confidence (log-prob, max-softmax)—our primary estimate. *cross*: train A vs. D (question fixed, only context changes), evaluate A vs. C. Within O=1 support and the synthetic factuality label coincide, so we make no distinctness claim.

5.2 No confidence asymmetry

We hypothesized that confidence absorbs the factuality axis more than the faithfulness axis, which would offer a way to separate them. The measured asymmetry is small and inconsistent in sign across the families and layers we examined (it does not favour the factuality axis), so we report a null result and do not rely on confidence-purging to distinguish the two notions.

5.3 Matched-stimulus contrasts align across families up to a linear map

We ask whether the support direction occupies the *same* geometry across families up to a linear map. The direct evidence is geometric: the Procrustes-mapped source direction is nearly collinear with the target family's *native* direction (transported-to-native cosine 0.976, range 0.95–0.99 over the

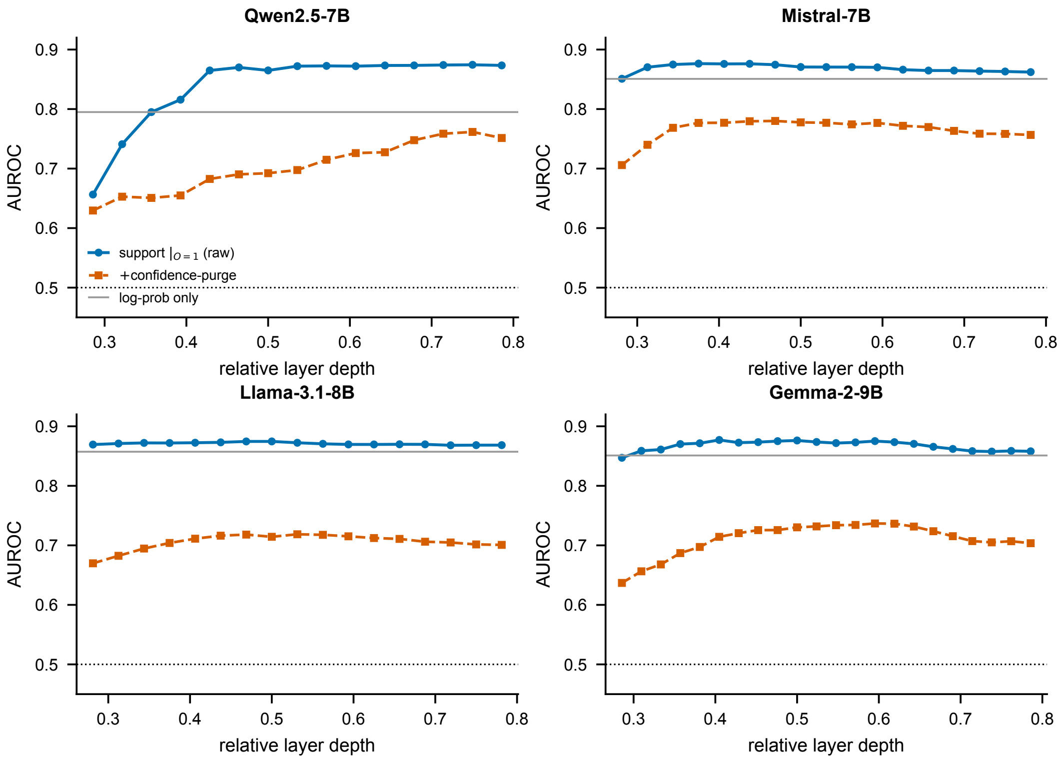

Figure 2: **The within-stratum contrast peaks mid-layer—but it is not grounding.** Cross-validated AUROC vs. relative layer depth for the support contrast within the answer-present stratum ($O{=}1$, where overlap is uninformative), the same after removing surface covariates, and factuality. The contrast is high and stable through mid layers on all four families (per-layer curves use all cells; the collision-free $\mathrm{supp}_{O=1}$ is in Table 1); a no-context ablation (§5.1) shows it is mostly question–answer compatibility, not context use.

12 ordered pairs; Table 2), well above anchor projection. The cosine is far above its null: a random or label-shuffled source direction transported through the *same* fitted map is essentially orthogonal to the native direction (cosine 0.01 and 0.10 respectively), so 0.976 is not an artifact of the shared-stimulus geometry. We deliberately do *not* headline the transferred-probe AUROC (0.906): it equals the within-family bound (0.905) and is therefore ceiling-saturated; scored with polarity-honest (signed) AUROC, random and shuffled-label directions through the map sit at chance (0.51/0.51). **This alignment is not specific to grounding.** Transporting the *factuality*, *answer-length*, and *lexical-overlap* directions through the same maps gives cosines of 0.99/0.98/0.99—as high as support—whereas a purely model-internal *confidence* direction transports at only 0.50 (and unstably, 0.04–0.90 across pairs). So what aligns is any contrast *derived from the shared input stimuli*, a shared low-rank stimulus geometry, not a grounding-specific axis; that confidence does *not* align shows the alignment is

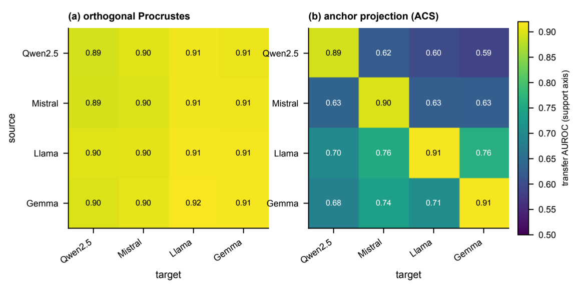

Figure 3: **Matched-stimulus contrasts align across families up to a linear map.** Source→target transfer AUROC of the support direction. (a) Orthogonal Procrustes is uniformly high on- and off-diagonal; we read this as alignment (cosine 0.98, Table 2), not as lossless transfer, since the AUROC is ceiling-saturated. (b) Anchor projection (ACS) matches it only within-family and degrades sharply across families.

non-trivial but also that it carries no grounding-specific content. We therefore conclude only that *matched-stimulus contrasts—support among them—are linearly alignable across these four 7–9B families*, and avoid any claim of a grounding-specific or near-lossless transfer.

axis	cosine	AUROC	within	rand-map	rand-lab
SUPPORT	**0.976**	0.906	0.905	0.519	0.506
FACTUALITY	0.989	0.915	0.911	—	—
nuisance directions:					
length	0.976	—	—	—	—
overlap	0.992	—	—	—	—
confidence	0.504	—	—	—	—

Table 2: **C3 (cosine-based, leakage-controlled).** Mean over the 12 cross-family ordered pairs, GroupKFold on ITEM_ID. *cosine*: transported-to-native cosine (the alignment evidence; non-saturating). A random / label-shuffled source direction through the *same* map has cosine only 0.01 / 0.10 with the native direction. *AUROC/within*: transferred-probe vs. within-family AUROC—equal, hence ceiling-saturated. *rand-map/rand-lab*: random / shuffled-label directions through the map, signed AUROC, at chance. Support, factuality, answer-length and overlap directions all align at ≈0.98, while a model-internal confidence direction does not (0.50): alignment is a generic property of contrasts derived from the shared stimuli, not a grounding-specific axis.

	label-free			NLI	overlap
model	synth-d_S	conf	in-dom†		
Qwen2.5	0.657	0.723	0.766	0.673	0.753
Llama	0.591	0.623	0.816	0.673	0.753
Gemma	0.541	0.686	0.800	0.673	0.753

Table 3: **C4 (leakage-controlled).** RAGTruth hallucination detection (AUROC; three families—Mistral's RAGTruth extraction OOM'd). The label-free synthetic support axis (synth-d_S, support polarity fixed from CtrlPairs, a-priori mid layer, scored without the test-fold flip) is below NLI, confidence, and lexical overlap, and for the weakest family (Gemma, 0.541) is barely above chance. †in-domain probe uses RAGTruth human labels (not label-free).

5.4 Synthetic axes do not transfer to real hallucinations

We learn d_S on CtrlPairs and apply it to the fixed RAGTruth responses (hidden states extracted through that same model; the responses are RAGTruth's own, so the NLI and overlap baselines are response-intrinsic and identical across the three models), predicting the human faithfulness label (Table 3). Here we keep the *support-polarity* sign fixed from CtrlPairs and score without the test-fold flip, because flipping per-test inflates a near-chance axis; CtrlPairs and RAGTruth share no items, so there is no leakage in this transfer. Scored honestly, the label-free synthetic axis reaches only 0.54–0.66 AUROC (and drops toward chance for the weakest family)—*below* NLI (0.673), model confidence (0.62–0.72), and plain lexical overlap (0.753), which is the single best predictor on RAGTruth. Removing the polarity flip can only lower these numbers, so the C4 negative is if anything understated by the original table. Thus on real data overlap still dominates, and even NLI does not escape it. A more sophisticated sentence-level NLI baseline (SummaC-style, max entailment over context sentences) does no better: pooled 0.614/0.664 and within-task macro 0.596/0.579 (mean/min), essentially tied with the whole-response NLI (0.582, within sampling noise) and still well below overlap (Table 4)—consistent with RAGTruth hallucinations often being abstractive or reasoning-level errors that survive sentence-by-sentence entailment, on which a small off-the-shelf NLI model has limited headroom. An *in-domain* probe (5-fold grouped-CV L_2-logistic on the same mid-layer states, target=human faithfulness label) does recover a usable signal: AUROC 0.77/0.82/0.80 for Qwen/Llama/Gemma (item-bootstrap 95% CIs in Table 3). It exceeds our NLI baseline (0.673) on all three families—though that baseline scores each multi-sentence response with a single whole-response entailment check and is conservative (§5); against the strong overlap baseline (0.753) it wins on Llama and Gemma (paired bootstrap $p<0.001$) but only ties on Qwen (+0.01, $p=0.17$). Residualizing lexical overlap out of the features first drops it to 0.60–0.66, so much of even this signal is overlap. A real support signal therefore exists in the internals, but it is a *different direction* from the synthetic axis and is recoverable only with in-domain human annotation.

6 Discussion and Limitations

Our results separate two questions usually merged. *Does* a non-overlap signal exist in the internals? Within the overlap-controlled stratum a contrast is decodable (0.76–0.88 purged) and aligns across four families up to a linear map—but a no-context ablation shows this within-stratum contrast is mostly *question–answer compatibility*, not context use, and the alignment is a generic matched-stimulus property (length and overlap directions align equally well). An apparent "grounding" signal thus survives an overlap control yet is still not grounding. *Is* that direction a deployable, label-free hallucination detector? No—it does not transfer to RAGTruth, where lexical overlap remains

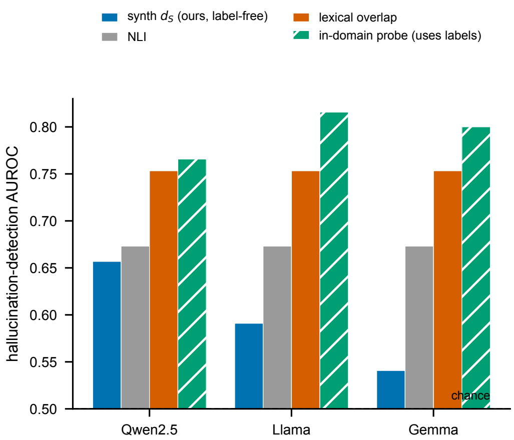

Figure 4: **The synthetic axis does not transfer to real hallucinations.** On RAGTruth, the label-free synthetic support axis (ours) sits below NLI and below plain lexical overlap, the strongest baseline; only an in-domain probe trained on human labels beats NLI. Axis starts at chance (0.5).

signal	Data2txt	QA	Summary	macro
synth-d_S	0.541	0.611	0.658	0.604
confidence	0.591	0.544	0.679	0.605
NLI (whole-resp.)	0.547	0.567	0.633	0.582
Sent-NLI (mean)	0.611	0.557	0.620	0.596
Sent-NLI (min)	0.521	0.591	0.626	0.579
overlap	0.605	0.686	0.649	0.647
in-domain†	0.742	0.721	0.679	0.714
faithful-rate	0.357	0.823	0.795	—

Table 4: **C4, per task type.** Within-task RAGTruth AUROC (synth-d_S, confidence, and in-domain are the mean over the three RAGTruth models; NLI and overlap are response-intrinsic). The pooled numbers in Table 3 overstate every signal because faithful base rates differ sharply across tasks (bottom row): overlap's pooled 0.753 falls to macro 0.647, NLI's 0.673 to 0.582. The *Sent-NLI* rows are the SummaC-style sentence-level baseline; they do no better (macro $0.596/0.579$). Within a task *every* label-free signal is weak (macro 0.58–0.65)—the synthetic axis (0.604) and confidence (0.605) included—and overlap (0.647) is merely the least-bad. †the in-domain probe (0.714 macro) uses RAGTruth human labels.

the strongest single-pass baseline (and even that edge is largely a between-task base-rate artifact: Table 4). The gap between a clean controlled signal and real-world utility is the main lesson, and it cautions against reading deployment claims from synthetic-stimulus probes. We are explicit about what we do *not* show: (i) within the answer-present stratum the decodable contrast is question–answer compatibility (a no-context ablation leaves it almost unchanged, raw 0.95–0.98) and coincides with our synthetic factuality label by construction, so we cannot claim a context-grounding signal there; only a question-fixed cross-strata control (0.66–0.88) shows any context-sensitive component; (ii) cell B has zero yield, so the fully-orthogonalized (corr$\approx$0) design is not instantiated and all analyses use cells A/C/D at corr$=$0.50 with overlap stratification; (iii) the cosine alignment is measured on shared stimuli, so it shows alignment of a shared low-rank contrast, not a fully unpaired transfer; (iv) RAGTruth covers three task types and one annotation scheme, the negative is on $3/4$ families (Mistral's RAGTruth extraction OOM'd), and stronger internals detectors (e.g. ReDeEP (Sun et al., 2025)) and sampling-based methods are out of our single-A100 budget—so we show that *this* synthetic axis fails, not that internals cannot work. Whether any label-free signal can decisively beat lexical overlap on real hallucinations is the natural next question.

Evaluation rigor. Because cells A/C/D of an item reuse the same context, all cross-validation and all Procrustes calibration/evaluation splits group on ITEM_ID (GroupKFold), so no near-duplicate rows cross the train/test boundary; estimates change negligibly from the ungrouped numbers (e.g. C1 $\text{supp}_{O=1}$ shifts by ≤ 0.01, C3 raw transfer stays $\approx$0.91), so the positive findings are not leakage artifacts. We resolve axis polarity on the train (or calibration) fold only; the test-fold $\max(\text{AUC}, 1-\text{AUC})$ convention floors a non-predictive direction above 0.5 (it inflates a true-chance random direction to ≈ 0.54–0.59), which matters for the *weak* numbers—the C3 map-aware nulls and the synthetic axis on RAGTruth—so we report those without the flip. The C1 analysis layer is fixed a priori at relative depth 0.5 (as C3/C4 already do) rather than argmax-selected on the reported metric. All confound-controlled estimates (confidence-purge, cross-strata, in-domain) and the C3 cosine and nulls are released as a single cached-feature script.

References

Shivam Adarsh, Maria Maistro, and Christina Lioma. 2026. How context shapes truth: Geometric transformations of statement-level truth representations in llms. *arXiv preprint arXiv:2601.06599*.

Amos Azaria and Tom Mitchell. 2023. The internal state of an llm knows when it's lying. In *Findings of EMNLP*, pages 967–976.

Waiss Azizian, Michael Kirchhof, Eugene Ndiaye, Louis Bethune, Michal Klein, Pierre Ablin, and Marco Cuturi. 2025. The geometries of truth are orthogonal across tasks. *arXiv preprint arXiv:2506.08572*.

Yuntai Bao, Xuhong Zhang, Tianyu Du, Xinkui Zhao, Zhengwen Feng, Hao Peng, and Jianwei Yin. 2025. Probing the geometry of truth: Consistency and generalization of truth directions in llms across logical transformations and question answering tasks. In *Findings of ACL*, pages 682–700.

Ivo Brink, Alexander Boer, and Dennis Ulmer. 2026. Probing for knowledge attribution in large language models. *arXiv preprint arXiv:2602.22787*.

Lennart Bürger, Fred A. Hamprecht, and Boaz Nadler. 2024. Truth is universal: Robust detection of lies in llms. In *NeurIPS*.

Seonglae Cho, Zekun Wu, Kleyton Da Costa, and Adriano Koshiyama. 2026. The confidence manifold: Geometric structure of correctness representations in language models. *arXiv preprint arXiv:2602.08159*.

Ekaterina Fadeeva, Aleksandr Rubashevskii, Dzianis Piatrashyn, Roman Vashurin, Shehzaad Dhuliawala, Artem Shelmanov, Timothy Baldwin, Preslav Nakov, Mrinmaya Sachan, and Maxim Panov. 2025. Faithfulness-aware uncertainty quantification for fact-checking the output of retrieval augmented generation. *arXiv preprint arXiv:2505.21072*.

Gemma Team, Morgane Riviere, and 1 others. 2024. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*.

Aaron Grattafiori and 1 others. 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Pengcheng He, Jianfeng Gao, and Weizhu Chen. 2023. Debertav3: Improving deberta using electra-style pre-training with gradient-disentangled embedding sharing. In *ICLR*.

Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Lélio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. 2023. Mistral 7b. *arXiv preprint arXiv:2310.06825*.

Su-Hyeon Kim and Yo-Sub Han. 2026. Cross-family universality of behavioral axes via anchor-projected representations. *arXiv preprint arXiv:2605.09875*.

Philippe Laban, Tobias Schnabel, Paul N. Bennett, and Marti A. Hearst. 2022. SummaC: Re-visiting NLI-based models for inconsistency detection in summarization. *Transactions of the Association for Computational Linguistics*, 10:163–177.

Moritz Laurer, Wouter van Atteveldt, Andreu Casas, and Kasper Welbers. 2024. Less annotating, more classifying: Addressing the data scarcity issue of supervised machine learning with deep transfer learning and bert-nli. *Political Analysis*, 32(1):84–100.

Shayne Longpre, Kartik Perisetla, Anthony Chen, Nikhil Ramesh, Chris DuBois, and Sameer Singh. 2021. Entity-based knowledge conflicts in question answering. In *EMNLP*, pages 7052–7063.

Samuel Marks and Max Tegmark. 2024. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. In *COLM*.

Ivan Vicente Moreno Cencerrado, Arnau Padres Masdemont, Anton Gonzalvez Hawthorne, David Demitri Africa, and Lorenzo Pacchiardi. 2025. No answer needed: Predicting llm answer accuracy from question-only linear probes. *arXiv preprint arXiv:2509.10625*.

Yixin Nie, Adina Williams, Emily Dinan, Mohit Bansal, Jason Weston, and Douwe Kiela. 2020. Adversarial NLI: A new benchmark for natural language understanding. In *ACL*, pages 4885–4901.

Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, KaShun Shum, Randy Zhong, Juntong Song, and Tong Zhang. 2024. Ragtruth: A hallucination corpus for developing trustworthy retrieval-augmented language models. In *ACL*, pages 10862–10878.

Charles O'Neill, Slava Chalnev, Chi Chi Zhao, Max Kirkby, and Mudith Jayasekara. 2025. A single direction of truth: An observer model's linear residual probe exposes and steers contextual hallucinations. *arXiv preprint arXiv:2507.23221*.

Bruno Puri, Jim Berend, Sebastian Lapuschkin, and Wojciech Samek. 2025. Atlas-alignment: Making interpretability transferable across language models. *arXiv preprint arXiv:2510.27413*.

Qwen Team, An Yang, and 1 others. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Peter H. Schönemann. 1966. A generalized solution of the orthogonal procrustes problem. *Psychometrika*, 31(1):1–10.

Zhongxiang Sun, Xiaoxue Zang, Kai Zheng, Jun Xu, Xiao Zhang, Weijie Yu, Yang Song, and Han Li. 2025. Redeep: Detecting hallucination in retrieval-augmented generation via mechanistic interpretability. In *ICLR*.

James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. 2018. FEVER: a large-scale dataset for fact extraction and VERification. In *NAACL-HLT*, pages 809–819.

Pengbo Wang, Chaozhuo Li, Chenxu Wang, Liwen Zheng, Litian Zhang, and Xi Zhang. 2025. One space to rule them all: Jointly mitigating factuality and faithfulness hallucinations in llms. In *NeurIPS*.

Adina Williams, Nikita Nangia, and Samuel R. Bowman. 2018. A broad-coverage challenge corpus for sentence understanding through inference. In *NAACL-HLT*, pages 1112–1122.

Yuheng Zha, Yichi Yang, Ruichen Li, and Zhiting Hu. 2023. Alignscore: Evaluating factual consistency with a unified alignment function. In *ACL*, pages 11328–11348.

Yu Zhao, Xiaotang Du, Giwon Hong, Aryo Pradipta Gema, Alessio Devoto, Hongru Wang, Xuanli He, Kam-Fai Wong, and Pasquale Minervini. 2024. Analysing the residual stream of language models under knowledge conflicts. In *NeurIPS Workshop on Foundation Model Interventions*.

A Implementation Details and Reproducibility

All numbers reproduce from cached features with `reproduce.sh`; extraction code, result JSONs, and built datasets are released. Below we fix every choice a reader would need to re-run the study.

Prompt and answer span. Activations are extracted from a raw template (no chat formatting): `"{context}\n\nQuestion: {question}\nAnswer:"`, to which the *asserted answer*—a teacher-forced gold surface string (NQ-Swap $a_{\text{true}}/a_{\text{swap}}$ for CtrlPairs; the model-generated response for RAGTruth)—is appended with a single leading space and no special tokens. Inputs are *left*-padded, so the answer occupies the final positions. The probe feature $\texttt{last}_L$ is the last-token residual state at layer L (under left padding, the final answer token); we also cache $\texttt{mean}_L$ (mean over the answer span) but all reported results use $\texttt{last}_L$.

Confidence scalars. Computed over the answer span only (logits shifted by one to predict the answer tokens): *mean log-prob* (mean gold-token log-probability), *mean max-softmax* (mean per-position top-1 probability), and *first max-softmax* (top-1 probability of the first answer token). The confidence confound set in the purge comprises three scalars: mean log-prob, mean max-softmax, and first max-softmax.

Lexical overlap. Two quantities, both case-folded. OVERLAP (continuous, `lexical_overlap`) is the fraction of whitespace-tokenized answer tokens that occur as a substring of the lowercased context; $\text{OVERLAP}_{\text{measured}}$ (binary, the cell label) is whether the *whole* stripped lowercased answer string is a substring of the lowercased context. A cell is dropped when its measured presence disagrees with its intended overlap label.

Construction and the missing cell B. CtrlPairs cells: A$=(q, c_{\text{org}}, a_{\text{true}})$; C$=(q', c_{\text{org}}, a_{\text{true}})$ with q' a donor question from another item; D$=(q, c_{\text{sub}}, a_{\text{true}})$ with an entity-swapped context. Cell B $=(q, \text{paraphrase}(c_{\text{org}}), a_{\text{true}})$ would be supported but answer-absent; it is produced only when a paraphraser is supplied and an item passes a *dual* filter—token-overlap of the answer with the rewrite ≤ 0.34 *and* NLI entailment of the answer by the rewrite ≥ 0.5. Greedy rewrites that remove the answer string tend to fall below the entailment threshold, while those that stay entailed tend to retain the answer tokens; the released build supplies no paraphraser, so cell B has zero yield and experiments use cells A/C/D.

RAGTruth labels. Each (context, response) row gets one binary response-level label: FAITHFUL=1 iff the dataset's `hallucination_labels_processed` dictionary sums to zero (no flagged span). We use no sentence-level mapping; the sentence-level SummaC baseline (§5) decomposes the response at *scoring* time only. The same 2,000 fixed responses are scored by every probe model (hidden states/confidence extracted by teacher-forcing each response through that model), which is why the NLI and lexical-overlap columns are identical across models. The whole-response NLI baseline forms its hypothesis by declarativizing each (question, answer) pair via a fixed template and scores the entailment probability with the context as premise; the sentence-level variant instead uses each response sentence as a hypothesis (§5).

Probing and cross-validation. Directions are mass-mean (difference of class means). Within-stratum decodability uses 5-fold GROUPKFOLD grouped by source item: for CtrlPairs the group is the NQ-Swap item, holding all of its cells out together; for the in-domain RAGTruth probe the group is the response id. Distinct RAGTruth responses can share a source passage, which our grouping does *not* hold out together—so the in-domain probe's 0.77–0.82 is, if anything, an optimistic ceiling. Confidence purging removes the component of the hidden states linearly predictable from the confound set before fitting. Cross-family transfer fits an orthogonal Procrustes map on shared stimuli at each model's a-priori layer (nearest cached layer to relative depth 0.5).

Compute. All activation extraction runs on a single A100-40G; per-model batch sizes are reduced for large-vocabulary models to avoid OOM (Qwen 4, Gemma 2, Llama 6, Mistral 16). All analyses, the sentence-level NLI baseline, and figures run on CPU/Apple-MPS without a GPU.